EEE 554 – Random Signal Theory

Homework Assignment #1

Assigned: 22 January 2026
Due: 10 February 2026

Reading (from Papoulis and Pillai):

Chapter 2; Chapter 4, sections 4-1 – 4.4; Chapter 5, sections 5-3 – 5-5

Work to hand in:

1. Problem 2-6, page 44.
2. Problem 2-7, page 44.
3. Problem 2-10, page 45.
4. Consider events A and B that are exhaustive in S with $P(B) = \alpha P(A)$ where $\alpha \geq 1$. If A and B are independent, show that $P(B) = 1$.
5. Problem 2-13, page 45.
6. Problem 4-8, page 120.
7. Define
$$h(x) = \begin{cases} c & \text{if } x \in (n, n + 2^{-(n+1)}), \quad n = 0, 1, 2 \ldots \\ 0 & \text{otherwise} \end{cases}$$
   (a) Find a value of c for which h is a PDF.
   (b) Assuming the value of c determined above, calculate $\mathsf{E}[x]$ for a RV $\mathbf{x}$ with $f_{\mathbf{x}}(x) = h(x)$.
   (c) Construct a PDF $f_{\mathbf{y}}$ such that $f_{\mathbf{y}}(y)$ takes on arbitrarily large values on each set $A_b = \{y \mid |y| > b\}$.
8. Problem 5-1, page 164.
9. The RV $\mathbf{x}$ has PDF
$$f_{\mathbf{x}}(x) = \begin{cases} c(a^2 - x^2) & |x| < a \\ 0 & \text{otherwise} \end{cases}$$
   (a) Find c in terms of a.
   (b) Calculate $P(\{-a < 4\mathbf{x} < 3a\})$.
   (c) Find the mean and variance of $\mathbf{x}$.
10. A RV $\mathbf{x}$ is gamma distributed. Calculate the moments $\mathsf{E}[x^k]$ for $k = 1, 2, \ldots$. [Hint: Use the characteristic function.]
11. (Computer problem) Use a random number generator (e.g., the Matlab `randn` function) to generate $10,000$ data values drawn independently from a normal distribution with mean zero and variance four. Similarly, obtain $10,000$ data drawn independently from a uniform distribution with mean zero and variance four (e.g., with the Matlab `rand` function).
    (a) Produce histograms of both data sets.
    (b) For each of the two distributions, calculate the probability that a datum falls between 1 and 2. Compare these theoretical figures to the actual percentages of data that fall in this range in each data set.
    (c) Perform a two-sample Kolmogorov-Smirnov goodness of fit test on the two data sets (e.g., using the Matlab function `kstest2` and interpret the result.